RL Environments for Finance

A technical report on reinforcement learning environments for financial decision-making.

AUTHORS

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Team

Sarang Balan

Sarang is a machine learning researcher and quant working at the intersection of AI systems and financial markets. He is a Research Associate at UCL's Centre for Artificial Intelligence, where he works on training small and large language models, and consults on agentic systems research for Facebook AI Research (FAIR) and Anthropic, as well as for the human-data companies Scale AI and Labelbox. He was previously research staff at the Sargent Centre for Process Systems Engineering, building power-trading systems with Shell, and worked at the University of Cambridge on machine-learning systems for modelling battery chemistries. He began on the markets side as a structured options desk quant at Three Comma Capital in Dubai. He holds a B.Eng. in Chemical Engineering and an M.Eng. in Engineering and Applied Mathematics.

Jean-Philippe Maltais

Jean-Philippe is a fixed income investment professional with over twenty years of experience across the riskier end of the capital structure, including leveraged loans, high yield, special situations, and private equity. He pairs top-down macro analysis with bottom-up fundamental due diligence to uncover deep-value opportunities. His career began in investment banking at Merrill Lynch (BAML) and Barclays Capital, before he moved into principal investing in leveraged loans, mezzanine, and private equity at Park Square Capital. He then built a credit research and portfolio-management track record at Lucror Analytics, New Amsterdam Capital, and Man GLG, followed by senior investment roles at the multi-strategy hedge funds Millennium and ExodusPoint, and most recently ran a long/short, event-driven credit strategy at Palm Lane Capital. He holds a Diplôme in Finance from HEC Paris, an M.Sc. in Applied Physics from Télécom Paris, and a B.Sc. in Physics from Université Laval.

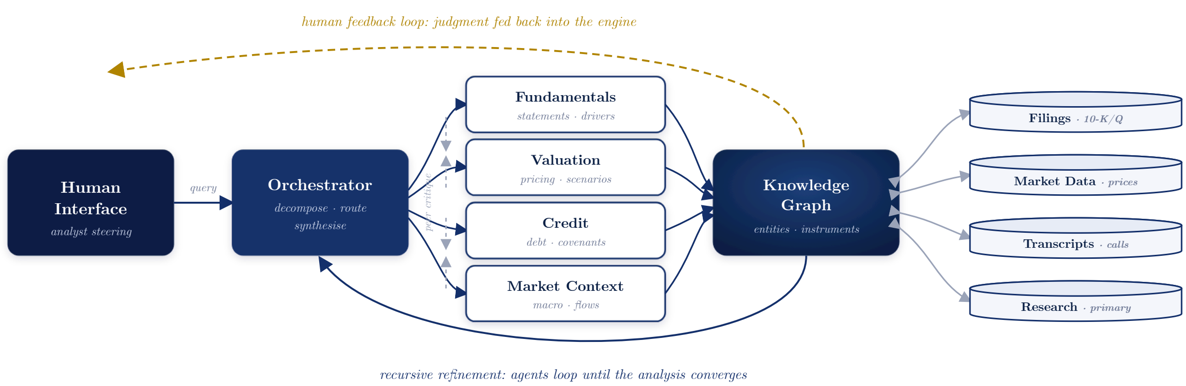
The Research Engine

The research engine is the synthesis of decades of financial experience: the judgment of practitioners who have priced risk through cycles, combined with an immense corpus of material gathered from the broadest possible range of sources, including regulatory filings, transcripts, market microstructure, credit documentation, and primary research. It is not a single model but an expression of those materials, their knowledge, and the techniques used to interrogate them.

Operationally, the engine is a **multi-agent system**. A central orchestrator decomposes a question, whether about a company, a sector, or a specific security, into focused tasks and dispatches them to specialised agents for fundamentals, valuation, credit, and market context. Each agent draws on dedicated data stores and a unified knowledge graph that links entities, instruments, and events, so that a claim about a stock is grounded in the same fabric as a claim about its debt. Agents critique and refine one another's output, and the system loops recursively until the analysis converges.

Crucially, the loop is closed by people. A human interface lets analysts steer the inquiry, challenge conclusions, and inject judgment that the corpus alone cannot supply, and every interaction feeds back into the engine, sharpening it over time. The result is a research capability that spans the full capital structure, from equities to debt, with the depth of an institution and the speed of software.

FIGURE 1 · ARCHITECTURE OF THE MULTI-AGENT RESEARCH ENGINE



Environment & Task Design

Each episode is a single corporate earnings event. The agent is given the prompt and a date-bounded RL sandbox of filings, transcripts, news, options chains and price history, and researches the name through open-ended tool calls. It returns one structured trade; the numeric fields of that output are scored against the realised post-earnings move to form a dense, continuous reward signal.

PROMPT

```
I want to analyze where and how {Aston Martin Lagonda} will
move for their upcoming earnings on Q{N} {YYYY}. The sandbox
contains filings, transcripts, news, options chains and price
history dated on or before {T}. Research the name and
return your analysis and a single trade as structured JSON.
```

RL SANDBOX

tools available: read_filing · search_news · get_options_chain · get_prices (data dated ≤ T)

ASSISTANT

Model thinking

Decompose the print: volumes & ASPs, guidance risk, net debt, options-implied move. Start with the filings.

read_filing

arguments:

```
read_filing("AML FY{YYYY-1} 10-K", sections=["MD&A","liquidity"])
```

result

```
revenue £1,633m · gross_margin 39.1% · net_debt £1.21bn · headroom thin
```

get_options_chain

arguments:

```
get_options_chain("AML.L", expiry ≈ earnings)
```

result

```
iv_30d 78% · straddle ⇒ implied_move ± 11.5%
```

⟳ reason ↔ act · peers (RACE, POAHY), news, scenario tests · … · synthesise → emit trade

STRUCTURED OUTPUT

```
{
  "thesis": "volume miss & FX drag vs. rich expectations",
  "financial_model": { "revenue_q": 415.0, "ebitda_q": 38.5 },
  "valuation": { "method": "EV/EBITDA", "fair_value_gbp": 1.05 },
  "position": "SHORT",          // ∈ {LONG, SHORT}
  "conviction": 0.62,           // 0-1
  "expected_move_pct": −8.4,    // post-earnings, T+1
  "expected_move_vol": 11.0     // forecast |move|
}
```

Bold numeric fields — direction, conviction and the move forecasts — are the reward-bearing predictions.

REWARD

```
realised move  r = (P_(t+1) − P_(t)) / P_(t)      // measured after the print
R = w_1·𝟙[sign(position)=sign(r)]              // directional hit
  + w_2·(1 − |expected_move − |r|| / s)         // magnitude accuracy
  + w_3·conviction·sign_match  −  penalties     // calibration − costs
```

A dense, continuous reward: directional accuracy, magnitude error against the realised move, and calibrated conviction.